

DCS MIGRATION AND SMOC UPGRADE PLANT1 Project.

YOKOGAWA

Project Delivery Department

Advance Process Control, Digital Enterprise Solution

February 2026

Content

PACE Operation HMI

APC : Advanced Process Control.

PACE : Platform for Advanced Control and Estimation.

หน้าต่าง Application (APP)

APC HMI - Platform for Advanced Control and Estimation - Run Time

Save | APC_ENGINEERS | Shed All Controllers | Help

Runtime Message | Message History

2/24/2026 10:27:17 AM Low Staged CPACE MAXAIR Processor CPACE: MAXAIR is unhealthy because mode is Inactive.
2/24/2026 10:27:17 AM Low Staged CPACE PXCO.PV Processor CPACE: PXCO.PV is unhealthy because status is BadPOV.
2/24/2026 10:27:17 AM Low Staged CPACE PXCO.PV Processor CPACE: PXCO.PV is unhealthy because status is BadMeas.

Navigation Pane

- Application Management
- Data Source Management
- Data Collection
- Lab Overview
- Custom Overview
- ML_4CBA
 - 4CBA
 - 4CBA_LR_VS
 - 4CBA_ML
 - 4CBA_ML_VS
 - PPACE[Staged]
 - PPACE
 - VS_PPACE
 - CPACE[Staged]
 - CPACE
 - Sub_CPACE
 - MAXAIR

Application Management

Application Name: CPACE | State: Staged | Actual Mode: ☒ On | Request Mode: On

Name	Description	Type	Status	Act Mode	Req Mode	Essential?	KPI (%)	Initial Req
CPACE	CPACE	Controller	NoOperSet	☒ Standby	Standby	False	0	None

ที่ตำแหน่ง APP จะแสดงตำแหน่งของ Application Management

ที่ตำแหน่งซ้ายมือตรง "App" สามารถ Drop down ได้ สะท้อน

- Main controller
- Sub controller

localhost : localhost | Engineer

เมนู Message History

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Application Name: CPACE | State: Staged | Actual Mode: On | Request Mode: On

Name	Description	Type	Status	Act Mode	Req Mode	Essential?	KPI (%)	Initial Req
CPACE	CPACE	Controller	NoOperSet	Standby	Standby	False	0	None

localhost : localhost | Engineer

ถ้ากดช่อง Message History จะแสดง Status ของ PACE ทั้งหมด

หน้าต่าง Main Controller (Ctr)

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Runtime Message

Time	Level	Status
2/24/2026 10:27:17 AM	Low	Sta
2/24/2026 10:27:17 AM	Low	Sta
2/24/2026 10:27:17 AM	Low	Sta

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Application Management

CPACE[Staged] | CPACE

Controller Name: CPACE | Status: NoOperSet | Actual Mode: Standby | Request Mode: Standby | KPI: 0 % | Tuning Status: Ok

Detail | Model Matrix | Auto Step Overview

Variable Set

Name	Description	Status	Act Mode	Req Mode	Essential?	KPI (%)
Sub_CPACE	Sub_CPACE	Inactive	Inactive	Inactive	False	0

Economic Function

Name	Description	Status	Act Mode	Req Mode	Value	St Priority
MAXAIR	NoMVOrCV	Inactive	Inactive	Active	NaN	9000

localhost : localhost | Engineer

Annotations:

- Show status ของ Main controller เช่น Off / Control / Simulate
- แสดง mode ของ Main controller OFF / Stand by / ON
- ปุ่มสำหรับ Operator Request ให้ Main controller OFF / Stand by / ON
- Show status ของ Sub controller เช่น OFF / Control / Simulate
- แสดง ACT Mode ของ Sub controller OFF / Stand by / ON
- แสดง REQ Mode ของ Sub controller OFF / Stand by / ON

Sub controller PTA

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Navigation Pane

Application Management PPACE[Staged] VS_PPACE

Application Management Data Source Management Data Collection Lab Overview Custom Overview

Variable Set Name Status Actual Mode Request Mode KPI Tuning Status

VS_PPACE Off Inactive Inactive 0% Ok

Control Estimation Standard Trends Custom Trends

MV

ค่าที่ APC จะ ปรับในอนาคด LO limit ค่า Move ที่ APC กำลัง ปรับ HI limit สถานะการ control Mode การ control Active / Inactive

Name	Description	Output	SS Value	LL	Move	HL	Unit	Status	Act Mode	Req Mode	SS Limit S
FC2102.SV	Solvent	48	48	120	0	130		Off	Inactive	Inactive	OP กด Request Active / Inactive
FC2201.SV	FC2201.SV	165	165	160	0	185		Off	Inactive	Inactive	
FC2101A.SV	FC2101A.SV	NaN	NaN	50	NaN	55		Off	Inactive	Inactive	

ค่าที่ APC ส่งไป control

ค่าปัจจุบัน ค่าที่ APC คาด การในอนาคต Setpoint control คือ ค่าที่ RTO ส่งมา สถานะการ control Mode การ control Active / Inactive

CV

Name	Description	Current Val...	SS Value	LL	Setpoint	HL	Unit	Status	Act Mode	Req Mode	SS Limit Stat
LI2101.PV	LI2101.PV	71.272	72.166	60	71.272	80		Off	Inactive	Inactive	OP กด Request Active / Inactive
AI2100Y.PV	AI2100Y.PV	1.088	1.088		1.088			Off	Inactive	Inactive	
FC2101A.CPV	FC2101A.CPV	NaN	NaN		52.5			Off	Inactive	Inactive	

LO limit HI limit

localhost : localhost Engineer

Sub controller CTA

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Navigation Pane

Application Management Application Management CPACE[Staged] CPACE Sub_CPACE

Variable Set Name Status Actual Mode Request Mode KPI Tuning Status

Sub_CPACE Inactive Inactive Inactive 0 % Ok

Control Estimation Standard Trends Custom Trends

MV

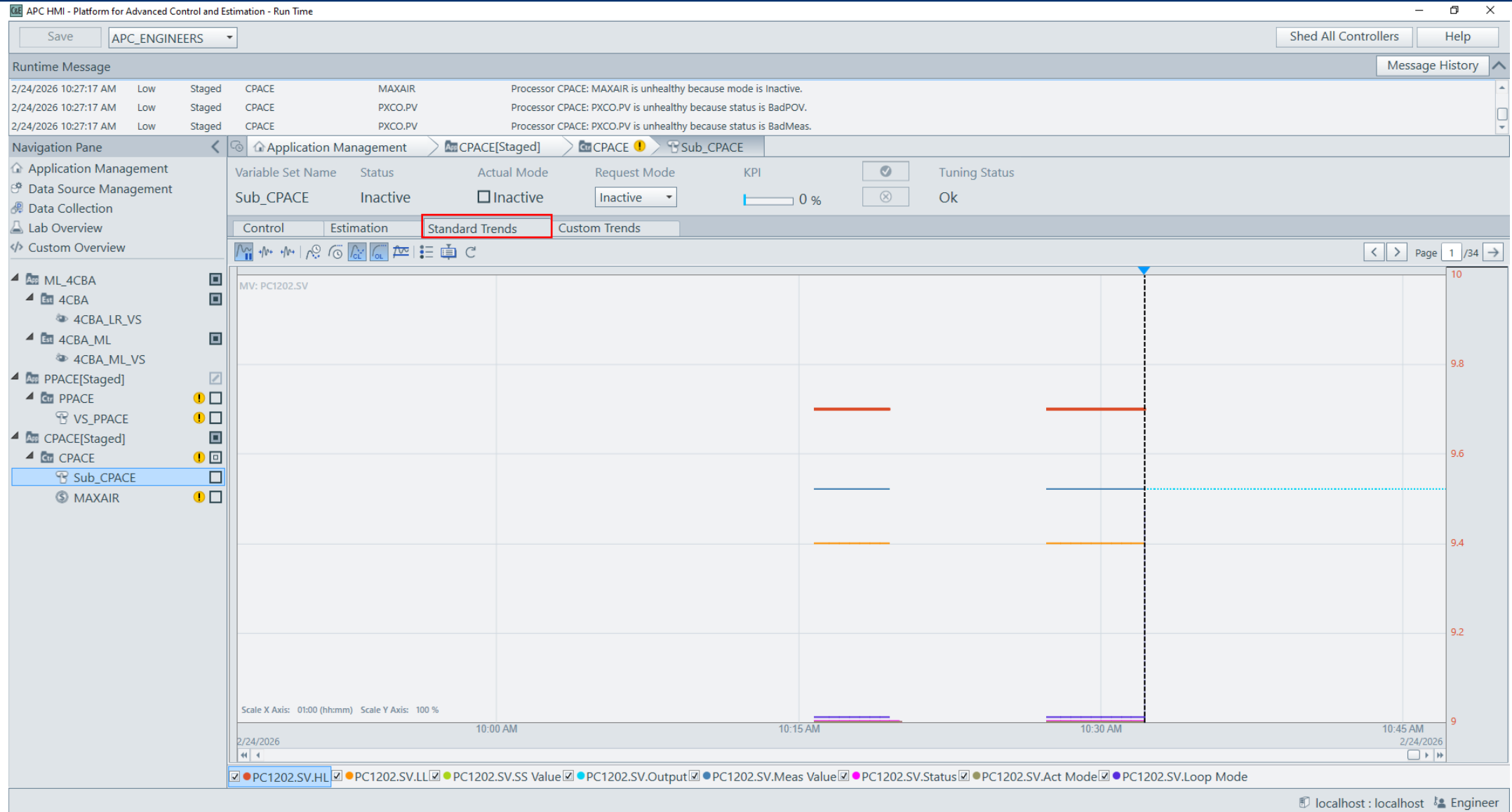
Name	Description	Output	SS Value	LL	Move	HL	Unit	Status	Act Mode	Req Mode	SS Limit S
PC1202.SV	PC1202.SV	9.523	9.523	9.4	0	9.7		NoOperSet	Inactive	Inactive	
FC1201.SV	FC1201.SV	0	0	134	0	138		InaptBLCMode	Inactive	Inactive	
FC1101.SV	TD-201	36.794	36.794	36	0	38		InaptBLCMode	Inactive	Inactive	
AC1104D.SV	AC1104D.SV	1056.357	1056.357	900	0	1200		NoOperSet	Inactive	Inactive	
FC1203Y.SV	FC1203Y.SV		16.572	16	0	18		InaptBLCMode	Inactive	Inactive	

CV

Name	Description	Current Val...	SS Value	LL	Setpoint	HL	Unit	Status	Act Mode	Req Mode	SS Limit S
PC1202.MV	PC1202.MV	0	9.654	0		100		NoOperSet	Inactive	Inactive	
PCT.CPV	MP	176.418	181.042	170		190		NoOperSet	Inactive	Inactive	
TAV1201Y.CPV	TAV1201Y.CPV	165.849	174.903	170		190		NoOperSet	Inactive	Inactive	
PXCO.PV	PXCO.PV	0	-6.567	16		17		NoOperSet	Inactive	Inactive	
AC1201.PV	AC1201.PV	0.327	0.327	2.8		3		NoOperSet	Inactive	Inactive	
PD1210.PV	PD1210.PV	-96.716	-96.716	500		800		NoOperSet	Inactive	Inactive	
PD1204.PV	PD1204.PV	-37.613	-37.613	170		250		NoOperSet	Inactive	Inactive	

localhost : localhost Engineer

ที่ Standard Trends สามารถดู trend ของการ control และค่าคาดการณ์ในอนาคต (Prediction value)



ที่ Custom Trends สามารถเลือก Trends ที่ต้องการดูได้

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Application Management

Variable Set Name Status Actual Mode Request Mode KPI Tuning Status

Sub_CPACE Inactive Inactive Inactive 0 % Ok

Control Estimation Standard Trends Custom Trends

Trend Pen Configuration

Source 1	Source 2	Source 3	Parameter Name	Display Low	Display High	Auto Scale	Common Scale
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐
CPACE	Sub_CPACE			0	100	☒	☐

Apply Cancel

Scale X Axis: 01:00 (hh:mm) Scale Y Axis: 100 %

10:00 AM 10:15 AM 10:30 AM 10:45 AM

2/24/2026

localhost : localhost Engineer

Symbol	Status
	Application status: On Processor mode: Active Variable Set mode: Active, Available Economic Function mode: Active
	Application status: Off Processor mode: Locked Variable Set mode: Locked, Unavailable Economic Function mode: Locked
	Processor mode: Inactive Variable Set mode: Inactive Economic Function mode: Inactive
	Processor mode: Standby Variable Set mode: Simulate Economic Function mode: Simulate

APC Controller : Active
APC Sub Controller : Active

APC Controller : Off
APC Sub Controller : Off

APC Controller : Inactive
APC Sub Controller : Inactive

APC Controller : Standby
APC Sub Controller : Simulate

Indicator	Name	Description
	OPC is bad quality	ค่า APC ของ MV CV DV OPV อ่านค่าจาก OPC ไม่ได้
	Unhealthy	แจ้งเตือนในกรณีต่างๆ เช่น ค่า MV CV และ HL LL
	ActiveHi	ค่า Current Value ของ CV MV และ HL
	ActiveLo	ค่า Current Value ของ CV MV และ LL
	ViolatedHi	ค่า SS Steady state prediction และ HL
	ViolatedLo	ค่า SS Steady state prediction และ LL
	Max Move is "ActiveUp"	APC MV กำลังปรับค่าขึ้น
	Max Move is "ActiveDown"	APC MV กำลังปรับค่าลง

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